Diffraction

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Explain spreading around edges and through openings.

SUCCESS CRITERIA - I can...

1. Compare opening size with wavelength.
2. Distinguish diffraction from refraction.
3. Use a wave picture to justify spreading.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “diffraction” and “wavelength” to explain your answer. Sentence starters are optional.

diffraction: wave spreading around an edge or through an opening

wavelength: distance between corresponding points on repeated waves

boundary: the place where a medium or constraint changes

01 Warm-up: bring back an earlier idea

A wave travels **9 m** in **0.3 s** at frequency **50 Hz**. **Find speed** and wavelength.

Word help: frequency = number of cycles each second

Sentence starter: I found … [unit]. This value means … in this situation.

Marzano check - circle: ① ② ③

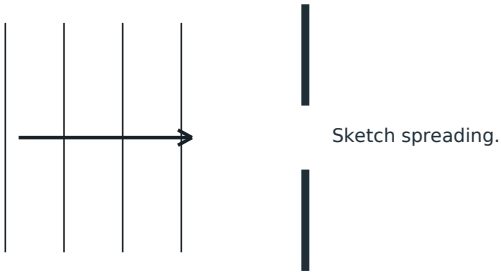
02 Worked example: explain the prediction

Reference: λ = v/f; **compare opening width** with λ. Spreading is pronounced when the opening width is comparable to wavelength. **Label the opening** and wavefronts.

Worked reasoning: A wave with wavelength **1 m** meets openings **1 m** and **10 m** wide. Spreading relative to opening size is more noticeable for the **1 m** opening.

Your turn: Draw incoming parallel wavefronts and the emerging patterns for each opening.

Sentence starter: In my work, I used … to show … . I labeled it … .

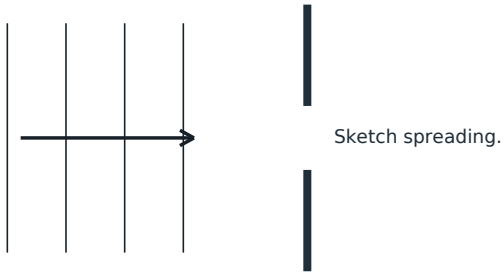


Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

1. **Sketch wavefronts after a narrow** opening about one wavelength wide.
2. Compare with a much wider opening.

Sentence starter: Both … . They differ in … because … .



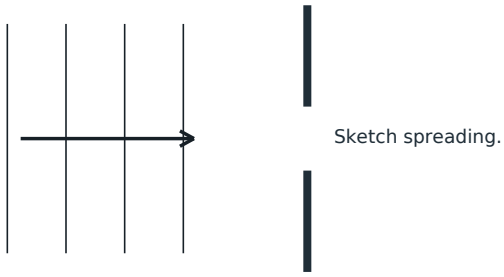
Marzano check - circle: ① ② ③

04 Apply the physics

A **340 Hz** sound travels at **340 m/s** toward a doorway **0.8 m** wide.

1. **Calculate its wavelength** and **compare wavelength**/opening ratios for this sound and visible light of wavelength **5.0×10^{-7} m**.
2. Use wavefront sketches to **explain why hearing around a** doorway differs from seeing around it.

Sentence starter: Both … . They differ in … because … .



Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Explain your reasoning

A student calls all wave bending “refraction.” Explain how diffraction can happen without crossing into a different-speed medium.

Sentence starter: Both … . They differ in … because … .

Marzano check - circle: 1 2 3

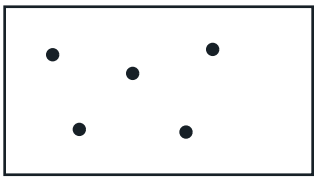
06 Review: connect an earlier idea

For a fixed ideal-gas particle mass, average translational kinetic energy becomes **16** times as large. **Compare rms speed**.
Use particle arrows and labels to **show the comparison**.

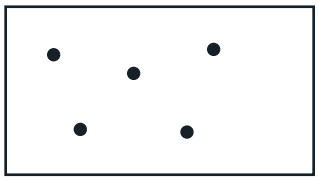
Word help: particle = a small constituent, such as an atom or molecule

Sentence starter: Both … . They differ in … because … .

Sample A



Sample B



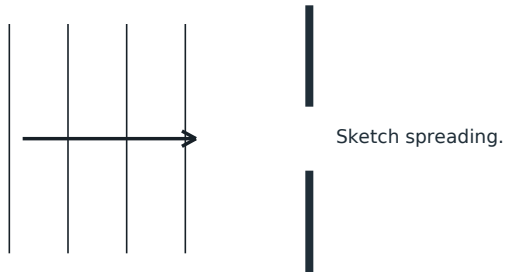
Add velocity arrows; use lengths to show a range of speeds.

Marzano check - circle: 1 2 3

07 Exit check

At fixed wave speed, frequency doubles. **Explain how** λ and typical diffraction through a fixed opening change.

Sentence starter: For this answer, I treated … as true. I found … . This supports … because … .

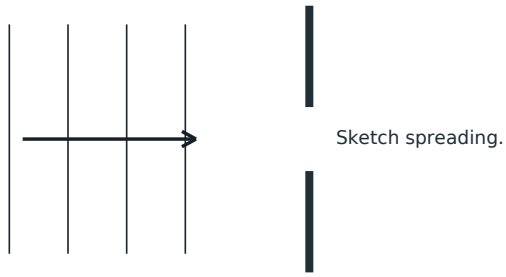


Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

Sound speed is **340 m/s**.
1. **Compare wavelengths** at **170** and **1700 Hz** with a **1 m** doorway.
2. **Explain which should diffract more** noticeably.

Sentence starter: Both … . They differ in … because … .



Marzano check - circle: 1 2 3